Project #2 - Project Plan (Final): Group 20

Title: Developing a High-Fidelity Hearing Protection Device to Ameliorate Heightened Sensitivity to Sounds for Patients with Hyperacusis and Tinnitus

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A. Problem Statement

Ringing in the ears, also known as tinnitus, is a medical condition that affects approximately 10-20% of the worldwide population [1]. For 10% of tinnitus sufferers, the condition can be so debilitating that it interferes with a person's ability to perform everyday tasks [2]. Current solutions to address problems associated with tinnitus are archaic or largely nonexistent. One of the main remedies that doctors prescribe to individuals diagnosed with tinnitus is to turn on a fan, or play white noise to drown out the ringing. If tinnitus persists, the patients are usually referred to a specialist to undergo cognitive behavioral therapy (CBT). CBT consists of exercises such as writing, journaling, and meditating, which are designed to change negative thoughts towards tinnitus, and to reduce anxiety and stress caused by the condition [1].

Current solutions have multiple shortcomings, as they do not tackle the underlying cause of tinnitus, nor do they address the persistent ringing caused by the condition [3]. Many tinnitus sufferers would love a solution that actually reduced the volume within their ears or, even better, cured the disorder completely. However, unfortunately, no such commercialized product currently exists and the issue is sadly, massively overlooked by the scientific community [4]. Moreover, when people acquire tinnitus from acoustic trauma resulting from loud noise exposure, oftentimes, hyperacusis, a heightened sensitivity to sound develops. People with hyperacusis tend to rely extensively on hearing protection, often in fear that the tinnitus will get worse [4]. This means patients are often required to wear such aids all the time, which is counterintuitive to current strategies like CBT that focus on improving tolerance to sound and loud noises. Conclusively, facing these challenges can be quite overwhelming for new tinnitus sufferers as they struggle with constant and simultaneous sound sensitivity.

As a result, we would like to design a solution for new tinnitus and hyperacusis patients to help them acclimate to the condition. A hearing protection device which modulates the exposure to high frequency sounds by 'smoothening' such wavelengths could serve as a potential prototype. Ultimately, the goal is to develop a high-functioning device that would relieve the daily suffering of patients with tinnitus by reducing harsh noises from their surroundings.

B. User Profile

One of our own group members, Andre Yu, has volunteered as the client for the project. He has been suffering from tinnitus and hyperacusis for over a year now, and is therefore, a suitable candidate for the role of primary stakeholder. While learning how to manage living with the condition in his day-to-day life, Andre has developed an extensive knowledge of common problems experienced by new tinnitus patients, and would thus be the right fit for our project.

C. Value Identification

Based on our initial research into individuals living with tinnitus and hyperacusis, several key personal values emerged that were deemed to be significant for this user group. People with tinnitus and hyperacusis often value having a sense of control over their environment and daily experiences. This primarily stems from the unpredictable nature of their condition and a strong desire to manage exposure to potentially triggering situations. Given the constant presence of internal sounds (tinnitus) and sensitivity to external ones (hyperacusis), individuals in this group typically place high value on moments of peace and the ability to find quiet spaces. This generally extends beyond mere silence, to encompass overall sensory comfort and harmony [5].

Many individuals suffering from tinnitus and hyperacusis value their independence, as well as their ability to participate in regular activities without constant reliance on others. For those who developed these conditions during their working years, maintaining their professional identity and career progression is highly valued, along with the ability to continue personal growth and development. Beyond managing their condition, individuals in this group typically value overall wellbeing, health, and their ability to manage their condition discreetly. Despite its challenges and struggles, patients diagnosed with this disorder are still quite eager to maintain meaningful social connections and relationships, while simultaneously holding the expectation that others would treat them with understanding, kindness, and respect, when needed [6].

Our proposed solution aims to embody these values by developing a system that helps users gradually become accustomed to living with the disorder on the regular, while maintaining their sense of control and independence. By providing users with tools to manage their sound exposure independently, track their progress, and make suitable adjustments to their environment according to their own comfort levels, we hope to support individuals with hyperacusis and tinnitus in achieving a better quality of life, while respecting their discretion and autonomy.

D. Empathy Development Plan
I. Stakeholder Questions

The following 5 questions have been compiled to better understand the stakeholder's perspectives and to conceptualize the social impact of the proposed design solution (*Table #1*).

Table #1: *List of Stakeholder Questions with Justification for Inclusion.*

No.	Questions	Justification
1.	Could you kindly share your experience living with tinnitus during its early stages and the type of condition you have?	This question aims to gather information on the type of tinnitus our stakeholder has, which differs in cause and symptoms, as well as their initial experiences upon being diagnosed with the condition [7].

2.	Could you please share events, activities, or places in your daily routine that would usually tend to worsen your symptoms?	As mentioned before, different individuals could have different triggers as well. We attempt to narrow the major causes of the stakeholder's symptoms that correspond the best to their current lifestyle.
3.	We understand that noise could be very distracting. Would it be possible for you to share your experience with different methods of symptom management and how effective they have been for you?	There are many ways one could manage the 'ringing' noise caused from tinnitus. However, not all treatments are suitable for all individuals that suffer from the condition [8]. Learning more about their experiences with managing their symptoms, as well as about what methods currently work the best, would allow us to design a device that suits their preferences.
4.	Are there any cons to your favorite method of tinnitus management? If so, would you mind letting us know about what these inconveniences are?	As a follow up to the last question, it would be great to understand minor inconveniences caused by their preferred treatment approach. Tinnitus is a condition which affects the user throughout the day. Hence, fixing these quibbles would greatly improve the effectiveness of the eventual design for our client.
5.	Are there any questions or concerns that we have not discussed yet that you would be comfortable sharing with us?	We invite questions at the end to let the stakeholder know that their concerns are valid and important for the eventual development of a tailored design.

II. Plan to Develop Empathy

Building an empathy-driven relationship is crucial to addressing the end user's needs and pain points. The team first explored empathy in inclusive design, which is defined as understanding and connecting with others' emotions to create meaningful relationships [9,10]. Striving for the "right" kind of empathy, the team aims to avoid over-identifying, which, as noted by the American Psychological Association, can misrepresent user needs [11]. The team will look to engage with the target user in a respectful and deliberate manner that reflects the desire to learn more about their challenges, related to tinnitus and current solutions.

To develop an empathy plan, the team held an initial meeting, for members to reflect on their empathy-related strengths and weaknesses, when working with individuals who have tinnitus. Some members had personal experience, bringing valuable insight, while most had little to no experience with the target group. To bridge this gap, the team will undergo a collaborative research and reflection process, by reviewing relevant resources which include but are not limited to; articles, blogs, dissertations and toolkits etc. The team will meet twice - early in the project (Oct 28 - Nov 1) and mid-project (Nov 11 - 15) - and, if meetings are not feasible, members will collaboratively annotate an online copy of the resource. This process aims to help us in understanding the needs of the target group and develop skills like culturally sensitive communication, while engaging with those who have tinnitus, to foster healthy connections.

With regards to the end user interviewee, the team will engage with them in a co-creation type manner, which traditionally involves involvement of the user throughout the design process [12]. To facilitate this continuous process, the team will collaborate consistently with the end user to develop a communication protocol. These may include bi-weekly progress reports for feedback, in addition to two user interviews: one at the start to guide development, and another near the end to gather valuable feedback. Interviews will be held at the user's convenience, in person or online. Using the lessons learned from the collaborative research initiative, individual research, and meetings with the target user, the team will iteratively reflect on the engagement process and identify possible improvements. If the user's participation is limited due to lack of availability or engagement with other commitments, the team will consult external stakeholders to inform decisions, including others with tinnitus, industry professionals, and researchers.

E. Work Plan

The following Gantt chart summarizes the major deadlines and work plan for all important stages of our project; which include Needs Assessment, Defining Specifications and Concept Generation (see Figure #1). Project Management is also included as a key component to ensure that communications with the client and TA progress in a timely fashion. This chart acts as a roadmap to streamline the design process and to ensure that all tasks are completed successfully within the given, three-month project timespan.

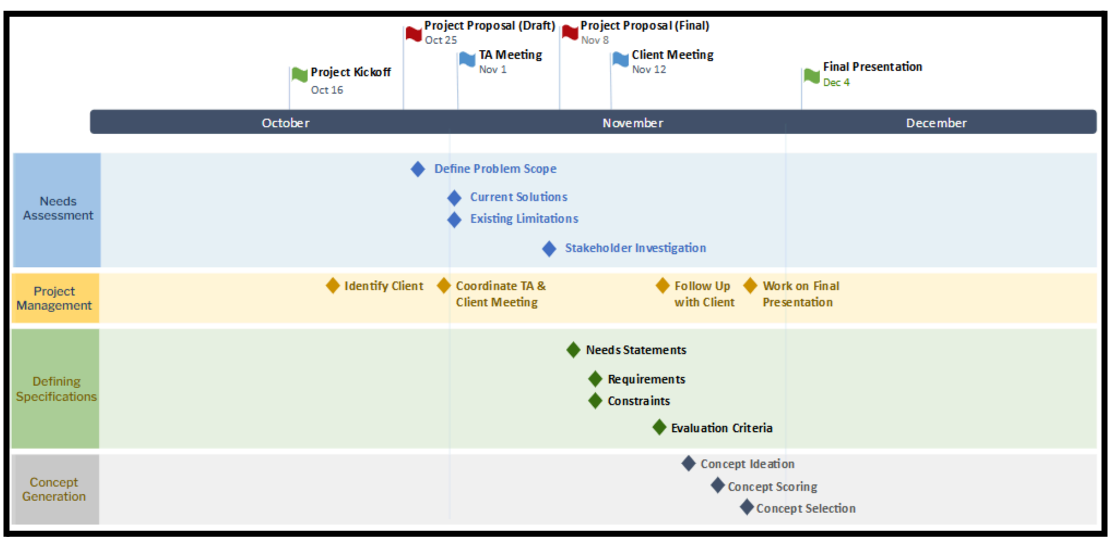

Figure #1: *Project Gantt Chart Across 3-Month Timeline.*

F. User Agreement Form

Course Overview: BMEG 455 – Professionalism and Ethics in Biomedical Design is a course that University of British Columbia (UBC) biomedical engineering undergraduate students take as a mandatory part of their professional degree program. The concepts taught in BMEG 455 focus on the fundamentals of what it means to be a professional engineer with emphasis on bioethics and equity, diversity, and inclusion.

Project Overview: The purpose of this project is for students to experience working on an inclusive design project and to gain perspective directly from users who would benefit most from the end product. The project will only cover the initial phases of a design project, focusing specifically on stakeholder engagement and ending at project specifications with 1-2 initial concepts described, but no completed final product. The project will not result in the construction or production of actual physical products. However, despite the lack of prototype development, students will get the opportunity to gather key insights from users, and understand how to identify needs directly from a user with a lived experience that is different from their own. The user also gets the opportunity to contribute ideas towards currently unaddressed or improperly addressed problems. This exposure will provide valuable training for budding engineers to develop skills in empathy and user/stakeholder needs identification, while utilizing inclusive language, so that as practicing engineers they will be more likely to confidently develop inclusive designs.

User Agreement: By signing below, I acknowledge that I have been informed of the nature and limitations of the BMEG 455 inclusive design project. I understand and agree to the following terms:

- I understand that the projects in this course are conceptual in nature and will not result in any actual physical product, construction, or implementation.
- I am comfortable with and am able to support the students (named below) by being interviewed for their project.
- I am comfortable with the students saving the following information from our interview to assist with their project (check all that apply):
  - Audio recording (✓)
  - Written transcript (✓)
  - Video recording (✓)
  - General written notes (✓)

Participant's Name: Andre Yu

Signature: Andre Yu

Date: November 8th, 2024

Group Members: Aly Khan Nuruddin, Albin Soni, Andre Yu, Jai Choraria, Pierce Alikusumah

G. References

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